Air Purifier Development

Market Research Analytics

Codebasics RPC *powered by* Dataful

AirPure Innovations

Suganya G

The Air Quality Challenge in India

India is home to 14 of the world’s top 20 most polluted cities.

Rising air pollution levels have triggered public health concerns and growing awareness.

AirPure Innovations aims to develop a smart, localized air purifier solution.

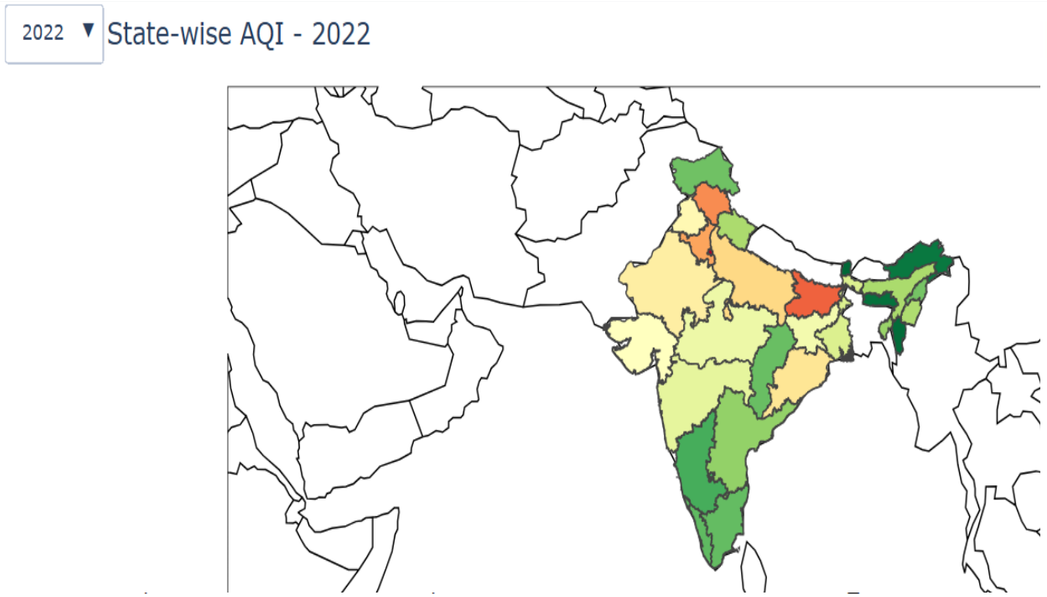
Defining the Path Forward

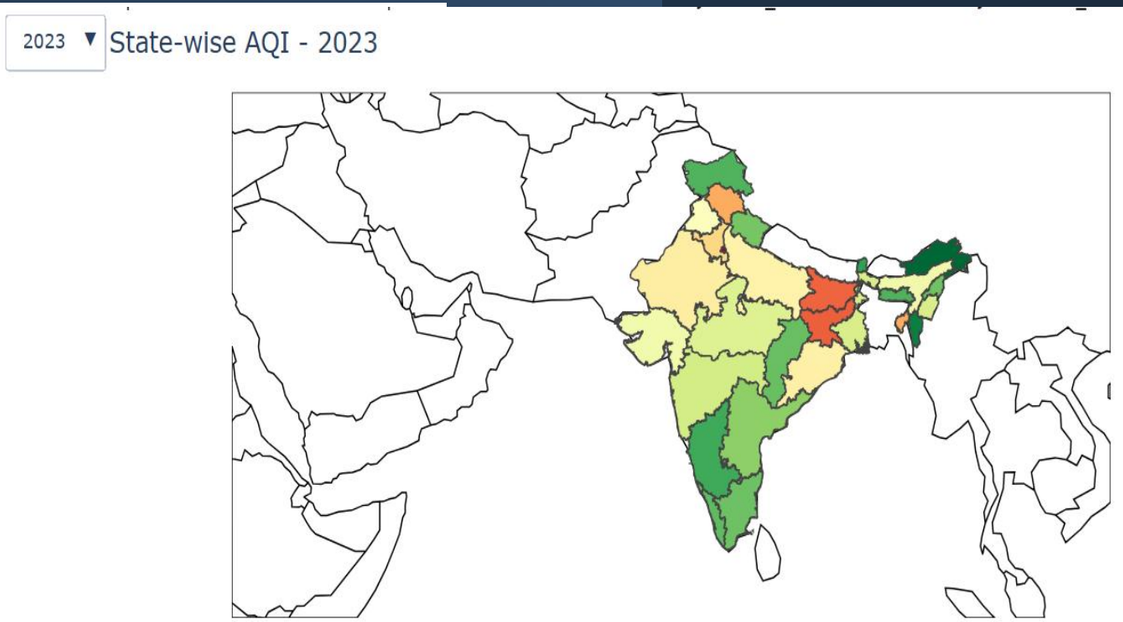
1. Target Pollutants & Particles
2. Essential Product Features
3. Market Demand & Opportunity
4. Aligning R&D with Localized Patterns

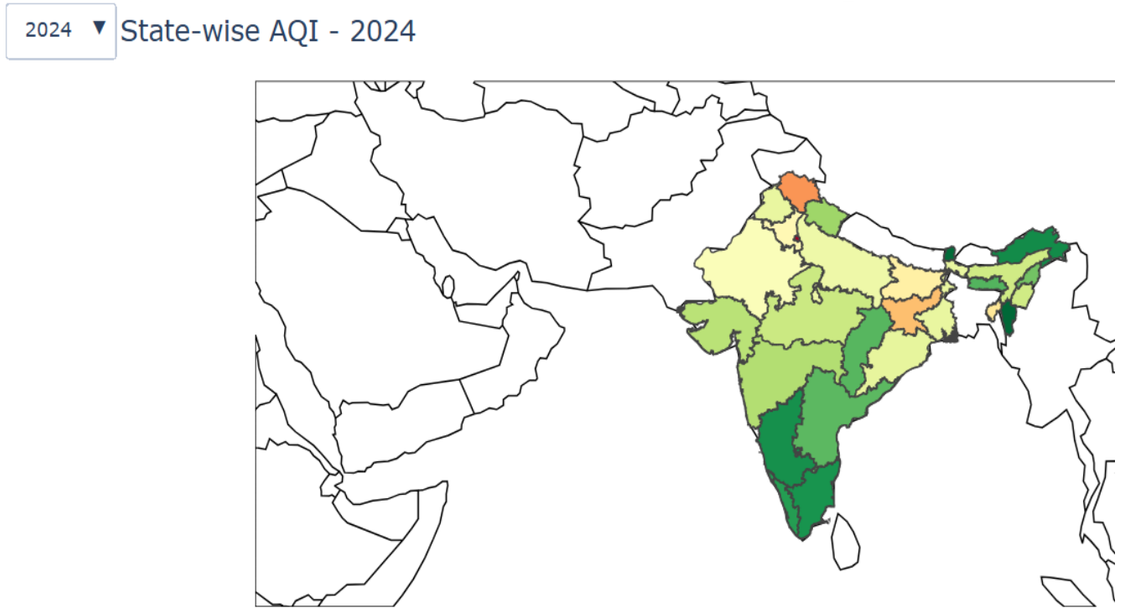
The Air We Breathe:
Understanding Air Pollution in India

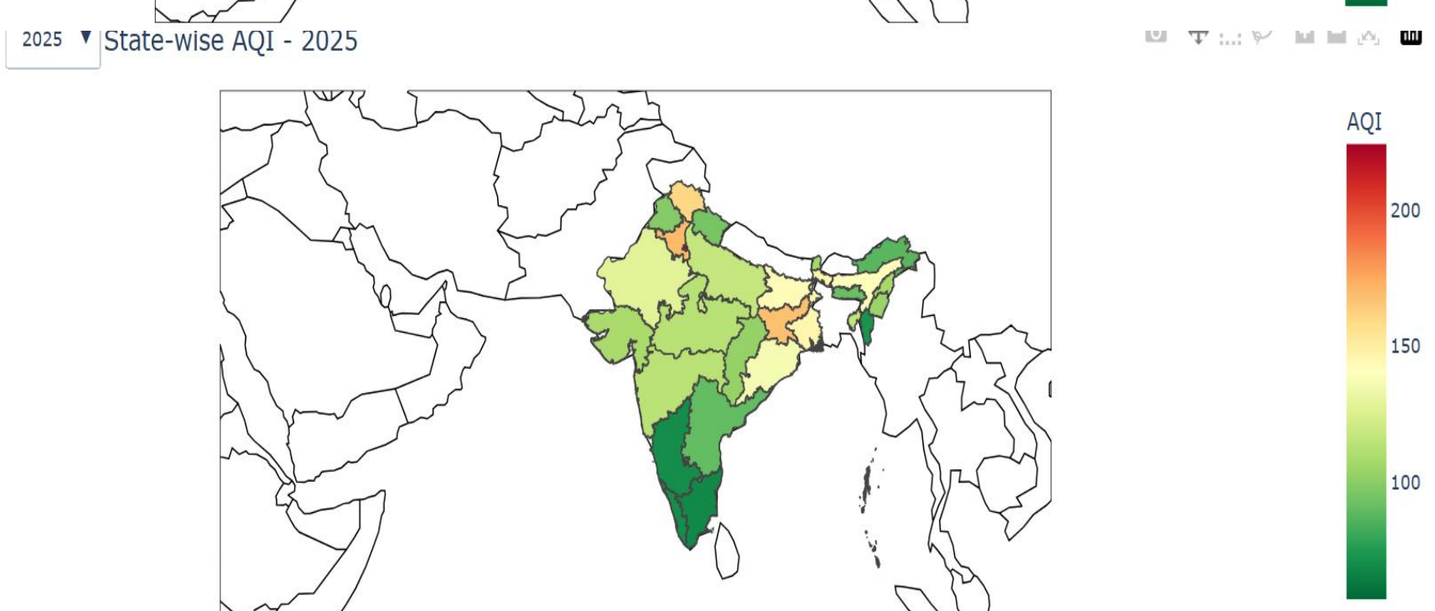


State wise AQI Trend

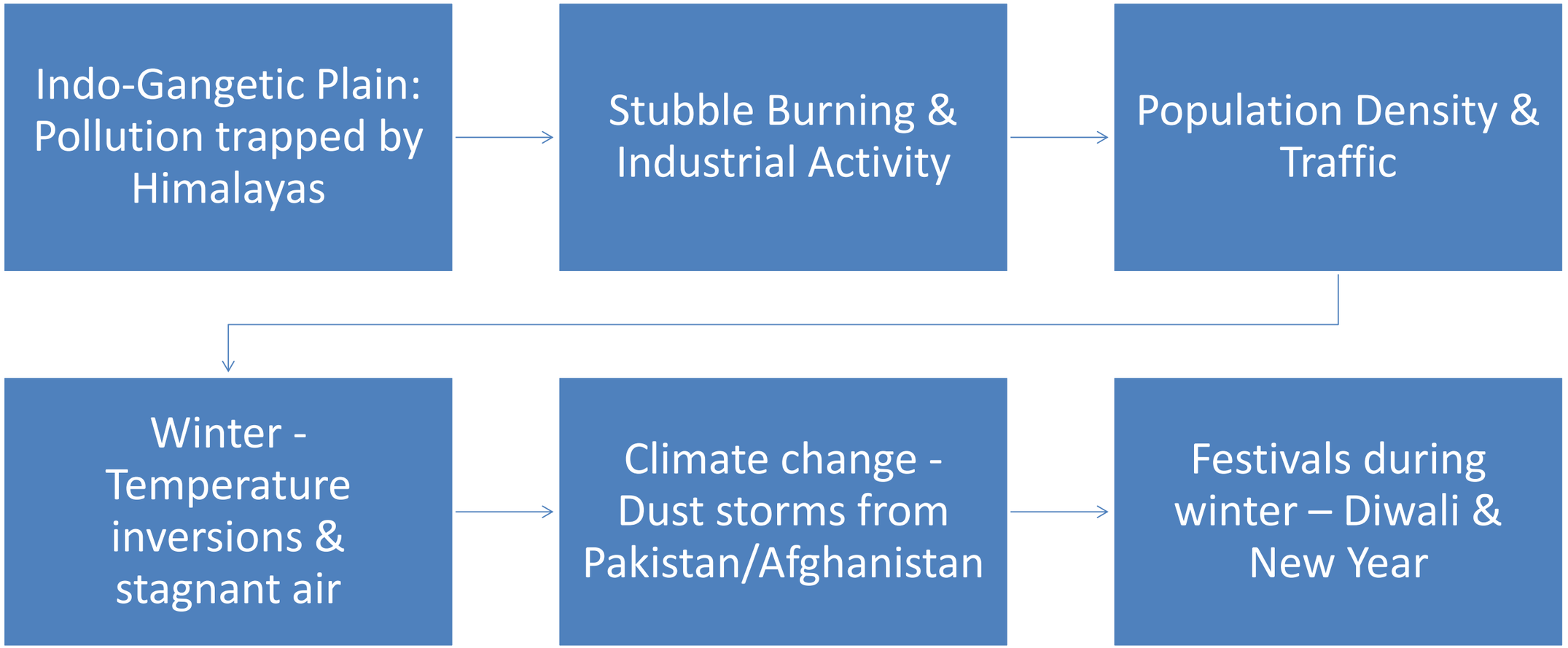




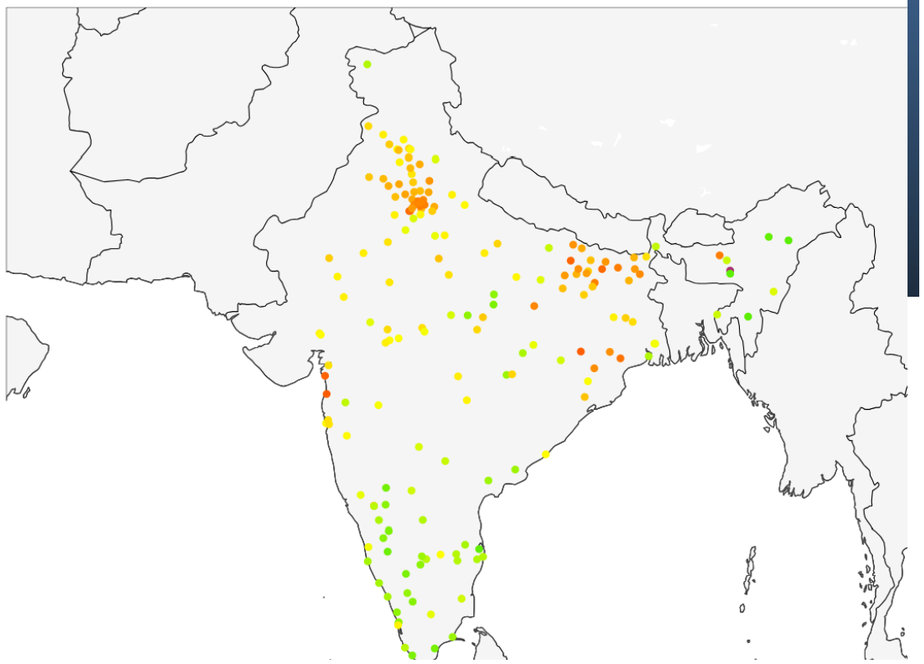




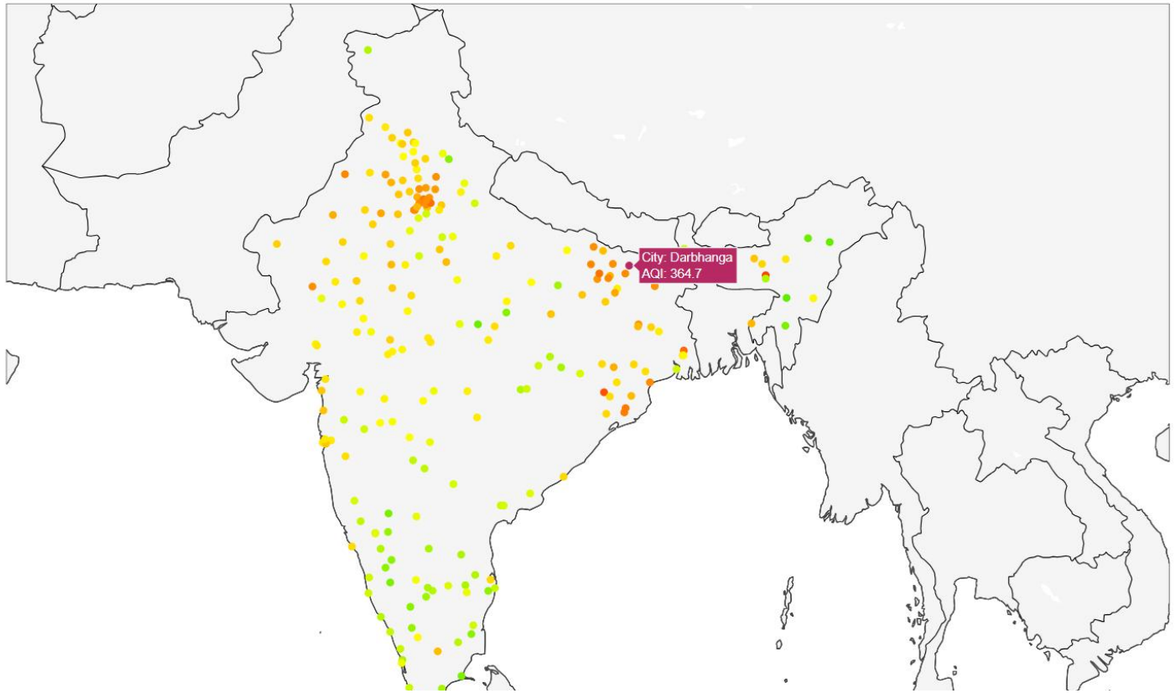
Why is North India affected more?

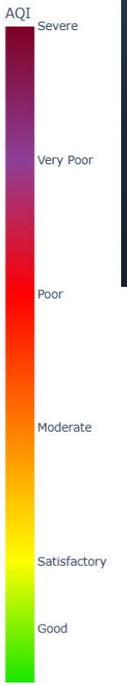


India AQI Map - 2022

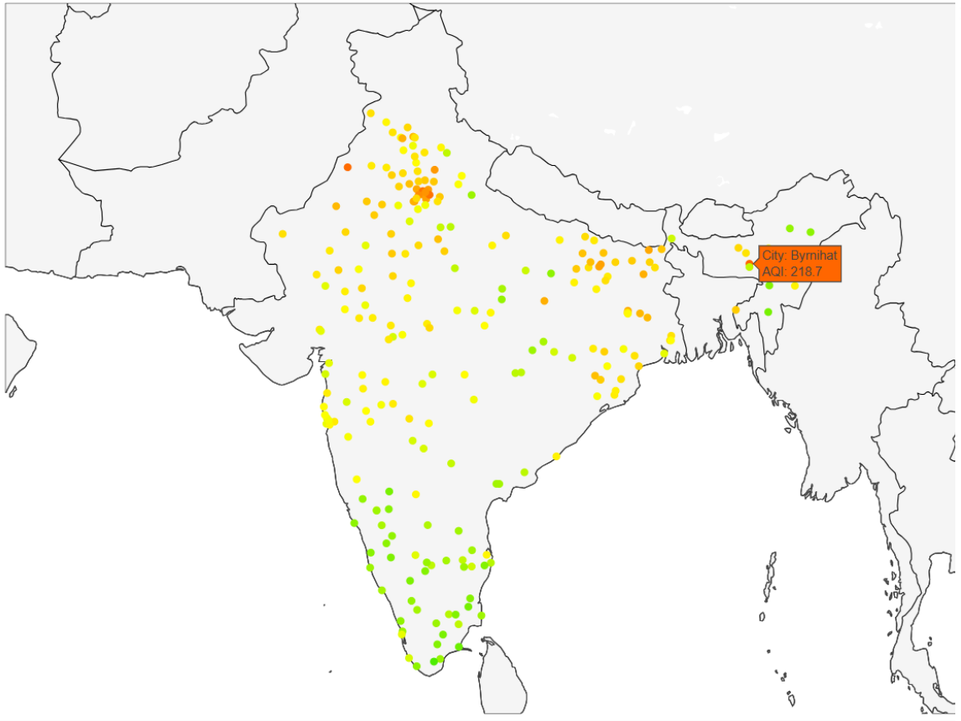


India AQI Map - 2023

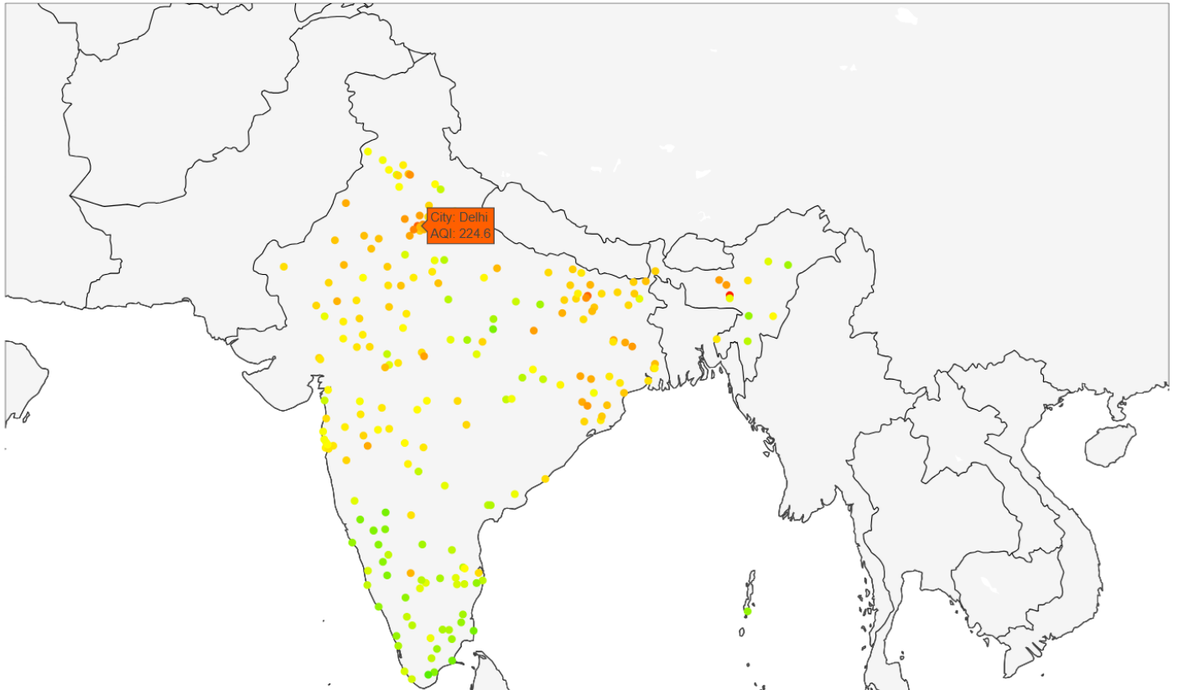




India AQI Map - 2024

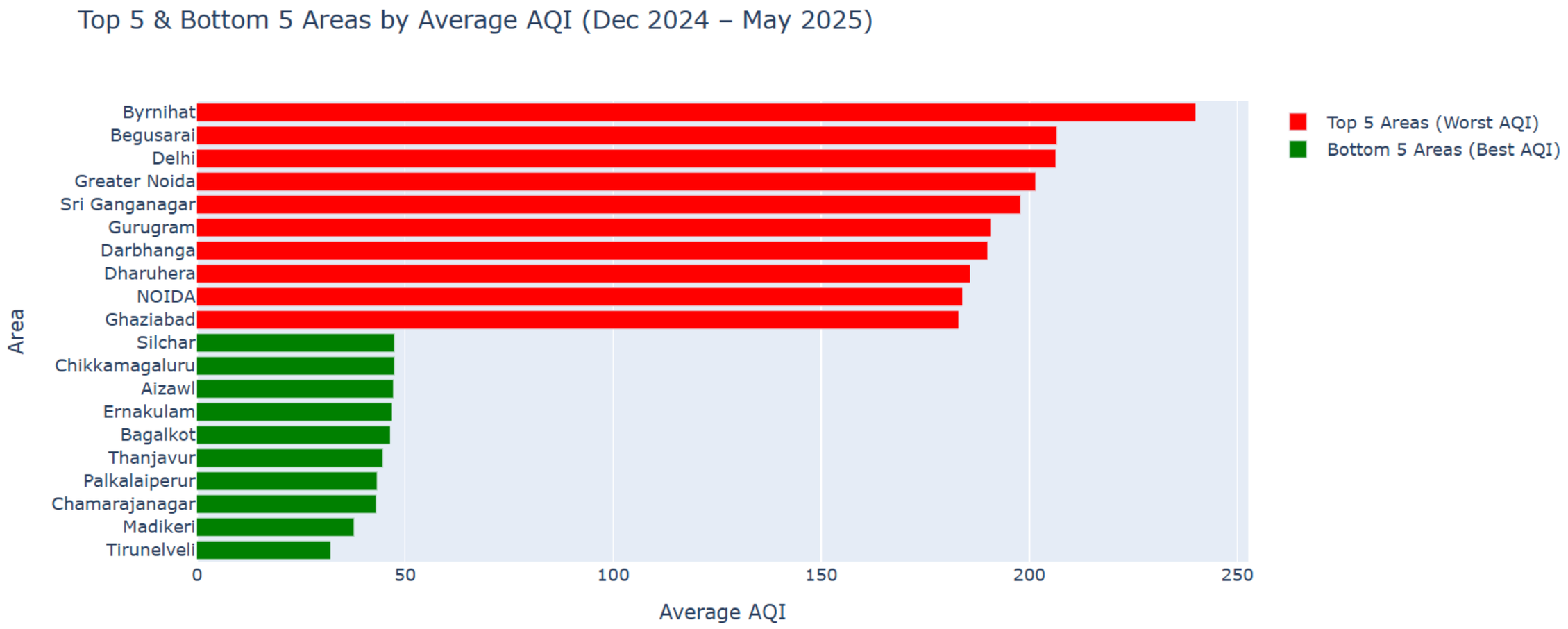


India AQI Map - 2025





Top & Bottom areas by average AQI *(Dec 2024 – May 2025)*

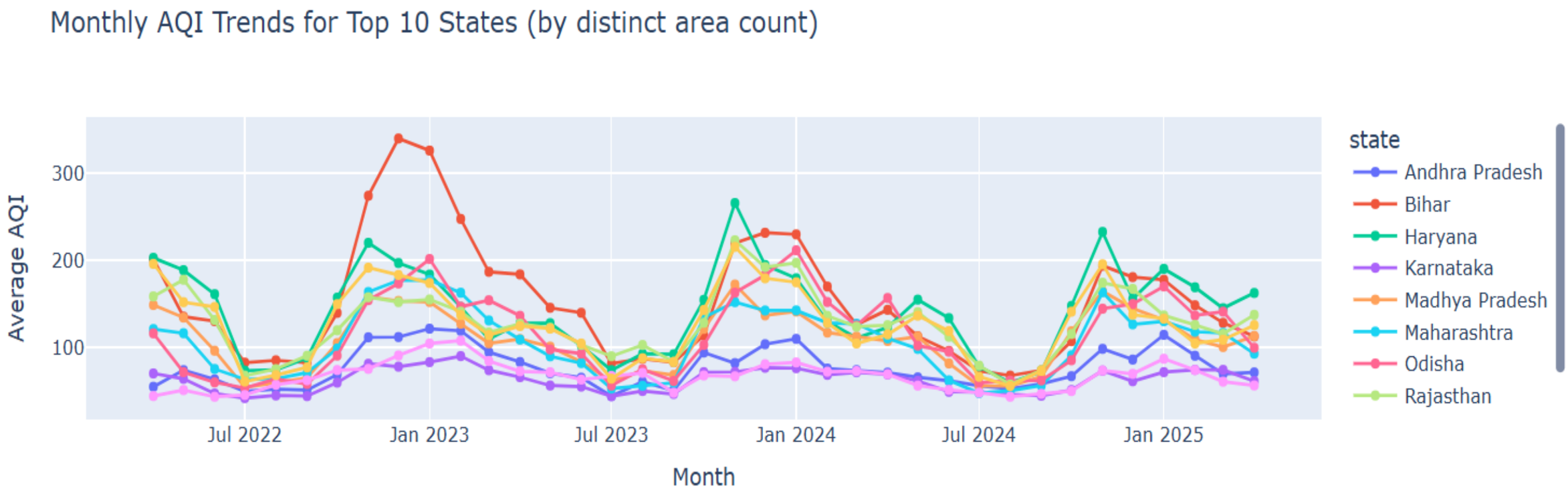


Top Pollutants across southern and northern states

state	top_1	top_2	top_3	top_4	bottom_1	bottom_2
Andhra Pradesh	PM10	PM2.5	O3	CO	NO2	SO2
Karnataka	PM10	CO	PM2.5	O3	NH3	SO3
Kerala	PM10	PM2.5	CO	NO2	NH3	SO2
Tamil Nadu	PM10	PM2.5	CO	SO2	NO2	NH3
Telangana	PM10	PM2.5	CO	O3	O3	NO2
Assam	PM10	PM2.5	CO	O3	NO2	SO3
Bihar	PM10	PM2.5	CO	O3	NO2	SO2
Delhi	PM2.5	PM10	O3	CO	CO	NO2
Haryana	PM2.5	PM10	O3	CO	SO2	NH3
Madhya Pradesh	PM10	PM2.5	O3	CO	SO2	SO3
Odisha	PM10	PM2.5	O3	CO	NO2	SO2
Punjab	PM10	PM2.5	CO	O3	NO2	SO2
Rajasthan	PM10	PM2.5	O3	CO	SO2	NH3
Uttar Pradesh	PM10	PM2.5	O3	CO	SO3	NH3
West Bengal	PM10	PM2.5	O3	NO2	SO2	NH3

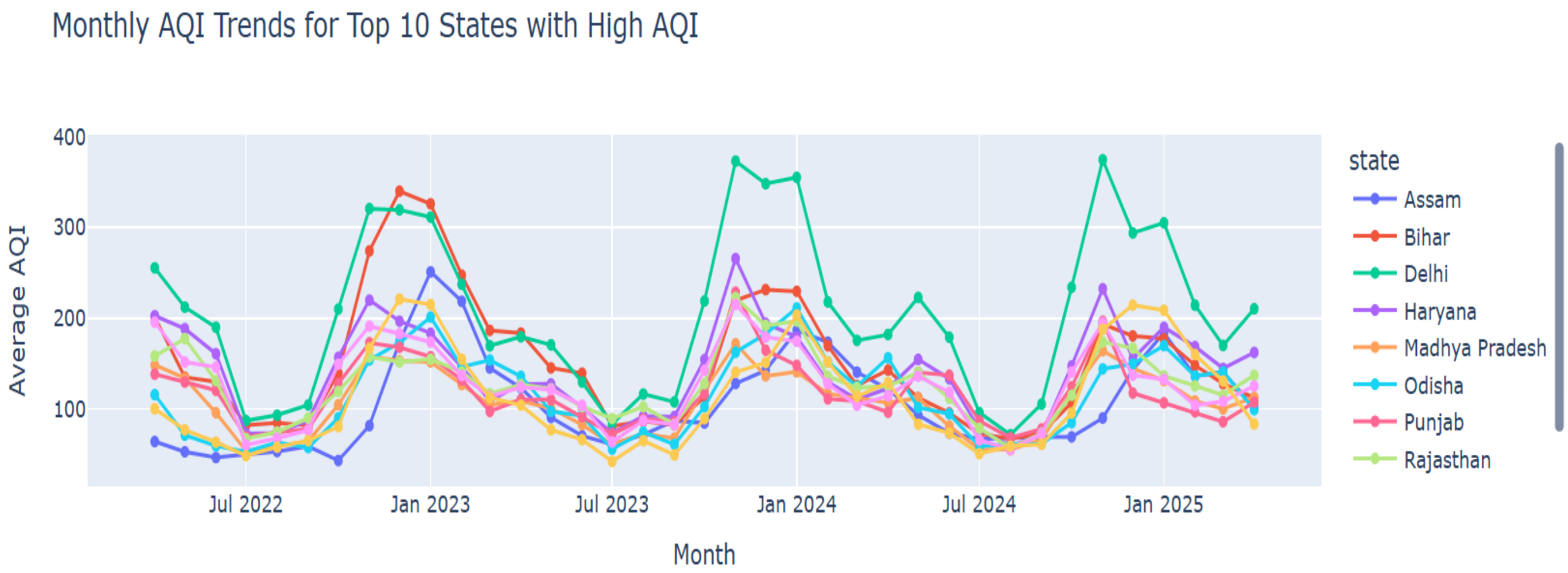
Monthly AQI Trends for Top 10 states with high distinct areas

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['Rajasthan', 'Maharashtra', 'Karnataka', 'Tamil Nadu', 'Bihar', 'Haryana', 'Uttar Pradesh', 'Odisha', 'Madhya Pradesh', 'Andhra Pradesh']
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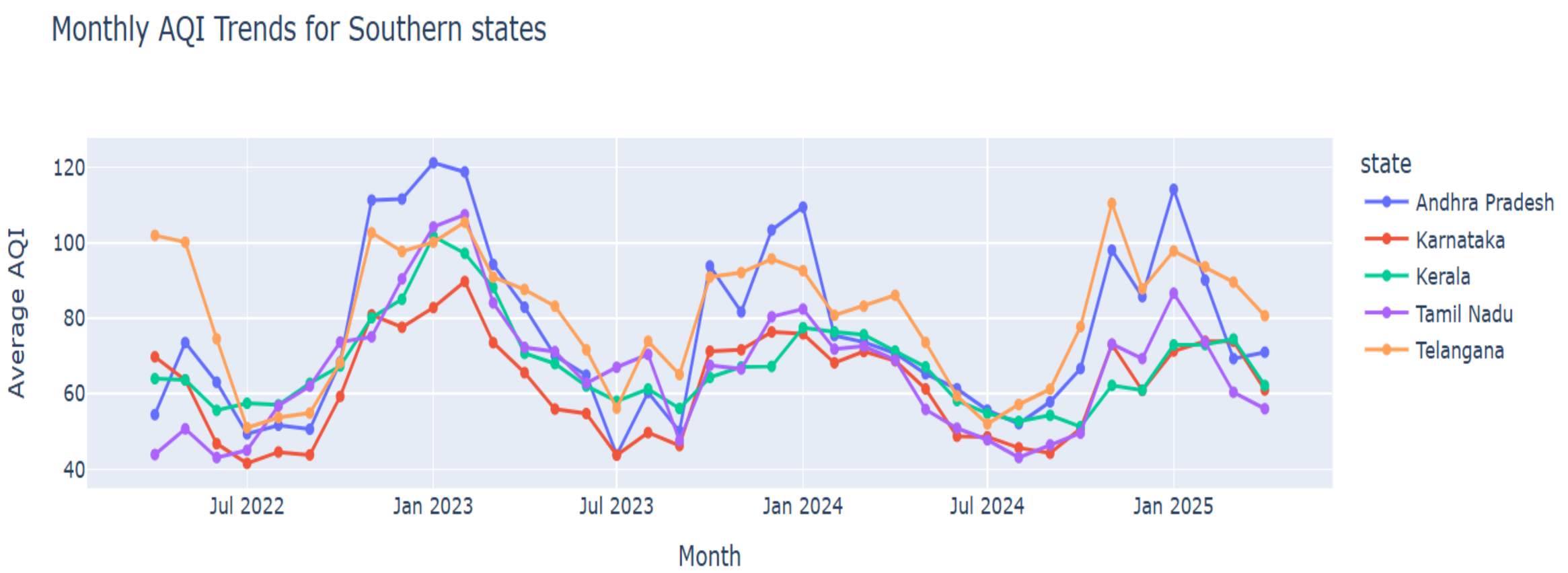
Monthly AQI Trends for Top 10 states with high AQI

['Bihar', 'Haryana', 'Uttar Pradesh', 'Rajasthan', 'Assam', 'Madhya Pradesh', 'Delhi', 'Odisha', 'Punjab', 'West Bengal']



Monthly AQI Trends for Southern States

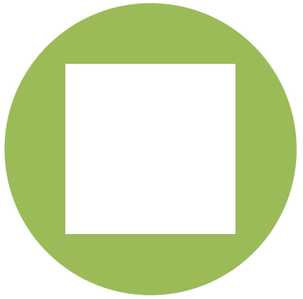
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['Tamil Nadu', 'Andhra Pradesh', 'Karnataka', 'Kerala', 'Telangana']
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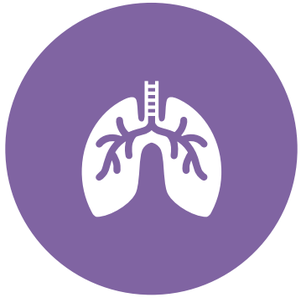
Seasons vs AQI trend



Monsoon (Jun–Sep): Clean air due to rains



Post-Monsoon (Oct–Nov): Crop burning & Diwali worsen air

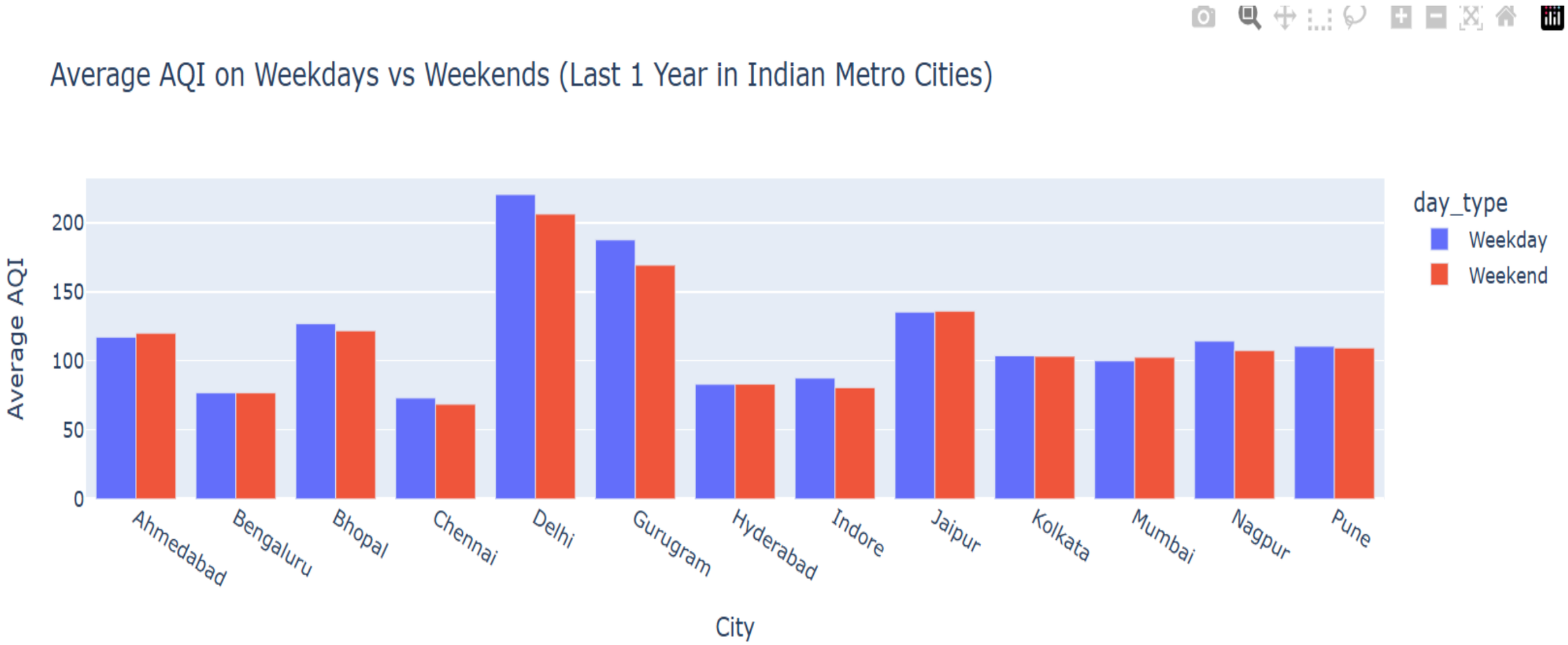


Winter (Nov–Feb): Trapped air and heating increase pollution



Summer: Dust storms in western India

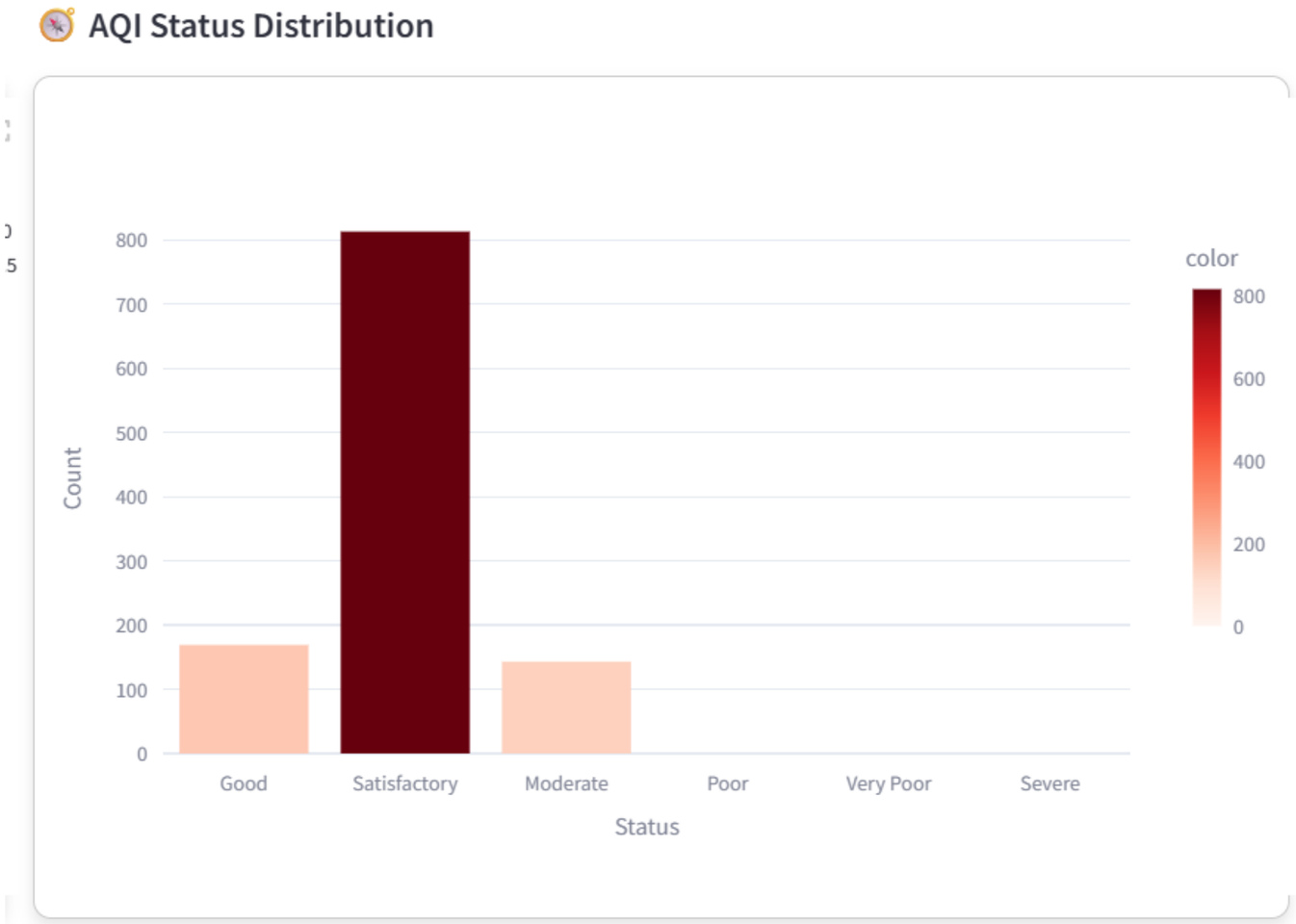
Weekend vs Weekday AQI Trend in Metro cities



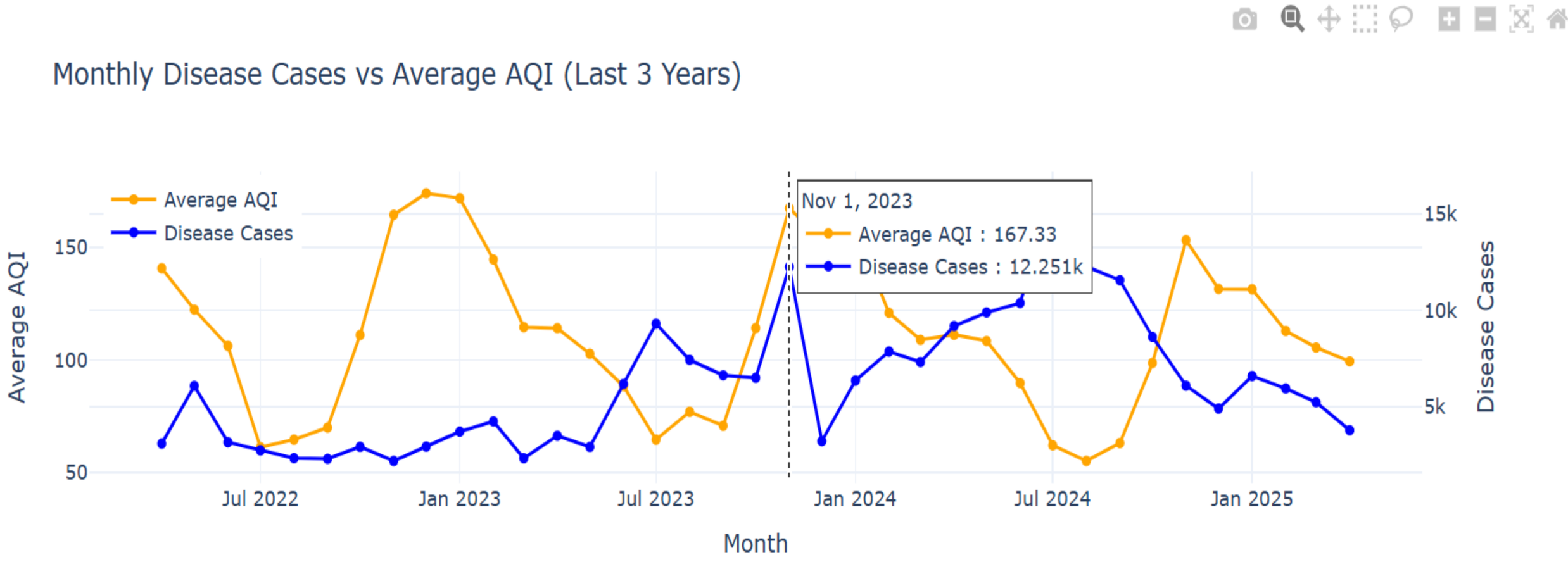
AQI status in Bengaluru between March-May 2025 & over past 3 years

(Major tech hub, rising construction dust)

Between March-May 2025	
air_quality_status	days
Satisfactory	48
Moderate	13



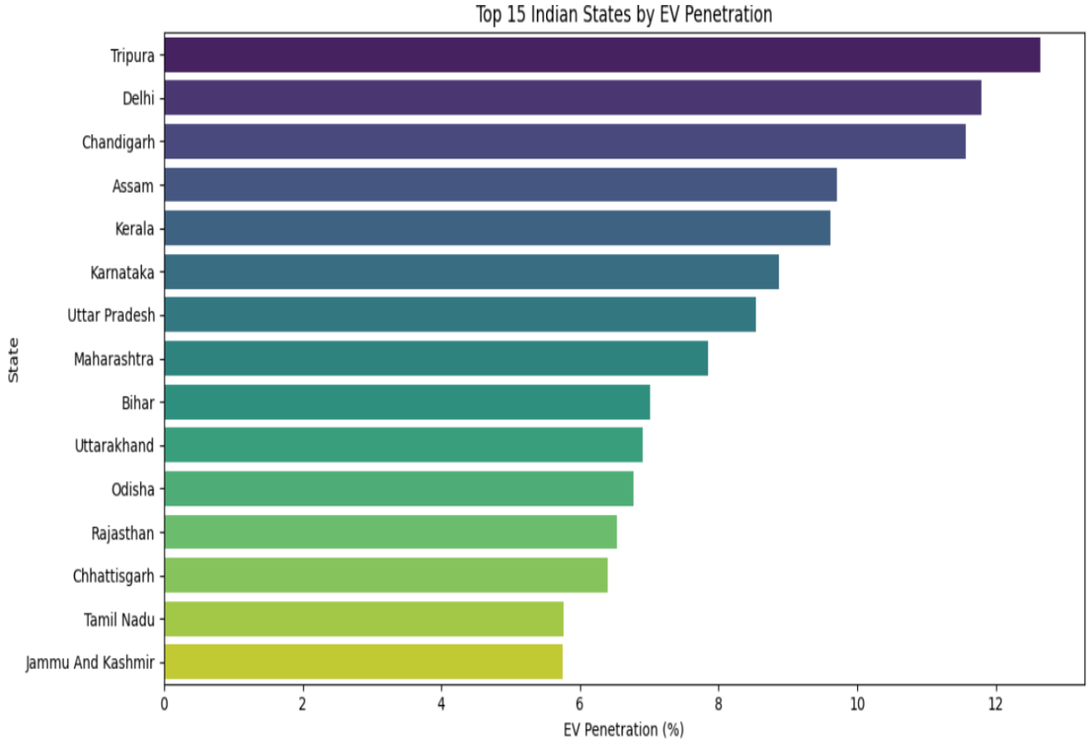
Diseases vs Average AQI over past 3 years



Top 2 diseases in each state in past 3 years vs AQI

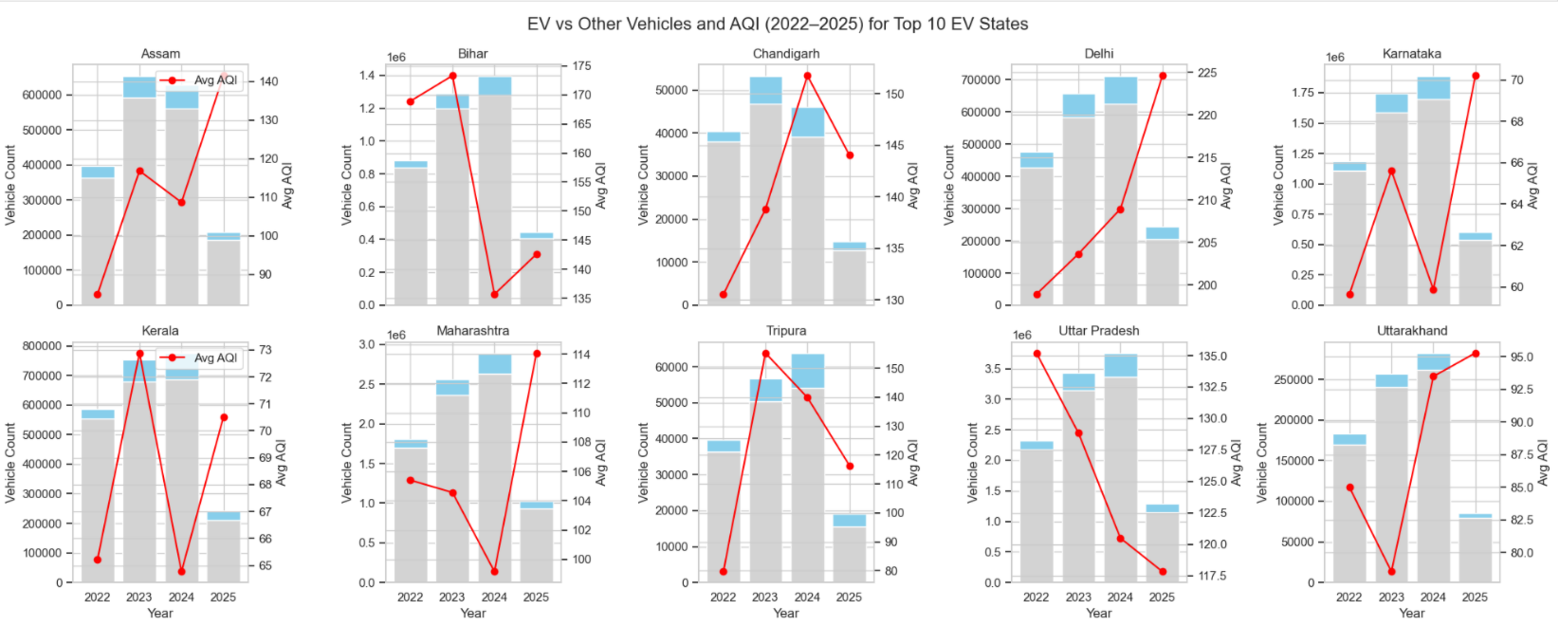
State	Disease1	Disease2	Cases1	Cases2	Avg_AQI
Andaman and Nicobar	Acute Diarrheal Disease	Fever with Rash	94	8	57.71
Andhra Pradesh	Acute Diarrheal Disease	Cholera	3189	1081	78.53
Arunachal Pradesh	Acute Diarrheal Disease	Mumps	290	133	55.96
Assam	Acute Diarrheal Disease	Food Poisoning	2683	1791	116.73
Bihar	Acute Diarrheal Disease	Fever with Rash	1316	830	153.26
Chandigarh	Cholera		16		145.12
Chhattisgarh	Acute Diarrheal Disease	Cholera	7818	745	79.58
Dadra and Nagar Have	Fever of Unknown andigin	Acute Diarrheal Diseas	101	72	
Daman and Diu	Food Poisoning	Acute Diarrheal Diseas	24	13	
Delhi	Dengue		40		208.85
Goa	Cholera	Food Poisoning	226	63	
Gujarat	Acute Diarrheal Disease	Cholera	4826	2287	109.19
Haryana	Cholera	Acute Diarrheal Diseas	972	696	136.86
Himachal Pradesh	Acute Diarrheal Disease	Hepatitis A	2181	559	161.01
Jammu and Kashmir	Dengue	Hepatitis A	1474	1140	69.71
Jharkhand	Malaria	Acute Diarrheal Diseas	5795	1609	165.85
Karnataka	Acute Diarrheal Disease	Cholera	6332	3616	63.51
Kerala	Food Poisoning	Acute Diarrheal Diseas	9342	4756	69.43
Ladakh	Mumps	Hand Foot and Mouth I	111	52	
Madhya Pradesh	Acute Diarrheal Disease	Dengue	5752	2207	108.76
Maharashtra	Acute Diarrheal Disease	Food Poisoning	6771	4382	103.44
Manipur	Dengue	Food Poisoning	3072	147	107.44
Meghalaya	Measles	Chickenpox	586	318	73.54
Mizoram	Food Poisoning	Scrub Typhus	339	233	51.52
Nagaland	Dengue	Acute Diarrheal Diseas	1244	72	83.43
Odisha	Acute Diarrheal Disease	Cholera	5495	3427	124.60
Puducherry	Acute Diarrheal Disease	Dengue	80	72	56.65
Punjab	Acute Diarrheal Disease	Cholera	889	715	117.09
Rajasthan	Acute Diarrheal Disease	Dengue	1185	836	128.22
Sikkim	Jaundice	Typhoid	136	111	59.20
Tamil Nadu	Acute Diarrheal Disease	Mumps	3028	1416	68.00
Telangana	Acute Diarrheal Disease	Food Poisoning	604	25	81.91
Tripura	Acute Diarrheal Disease	Dengue	390	259	143.62
Uttar Pradesh	Acute Diarrheal Disease	Food Poisoning	4140	2218	123.69
Uttarakhand	Dengue	Acute Diarrheal Diseas	1092	626	88.10
West Bengal	Acute Diarrheal Disease	Food Poisoning	3276	2360	117.82

EV Adoption Analysis



state	total_ev_vehicles	total_vehicles	ev_percentage	avg_aqi
Tripura	22631	178931	12.6	126.70
Delhi	245938	2086374	11.8	206.42
Chandigarh	17888	154562	11.6	141.56
Assam	183428	1889297	9.7	114.12
Kerala	226600	2357315	9.6	68.07
Karnataka	480191	5413304	8.9	62.67
Uttar Pradesh	921471	10782843	8.5	126.42
Maharashtra	650823	8287915	7.9	103.64
Bihar	280947	4009485	7.0	157.16
Uttarakhand	55704	806947	6.9	87.90
Odisha	153213	2260339	6.8	124.88
Rajasthan	305605	4679976	6.5	127.99
Chhattisgarh	117666	1837962	6.4	78.99
Tamil Nadu	329634	5725062	5.8	67.76
Jammu And Kashmir	30785	535491	5.7	69.50

EV Adoption Analysis – Top 10 states



Health Impacts in age groups due to poor AQI

Age Group	Health Impacts
Newborns & Infants	Elevated neonatal and infant mortality; low birth weight & preterm birth; developmental impacts
Children under 5 years	Respiratory illness (ARI, pneumonia), stunting, anemia, cognitive delays
School-age & Young Adults (18–30)	Respiratory symptoms, missed work/school, early cardiovascular stress
Middle-aged & 45–59	Mild cognitive impairment associated with biomass/indoor smoke exposure
Older Adults & Elderly (60+)	Increased frailty risk, accelerated cognitive decline, cardiovascular mortality

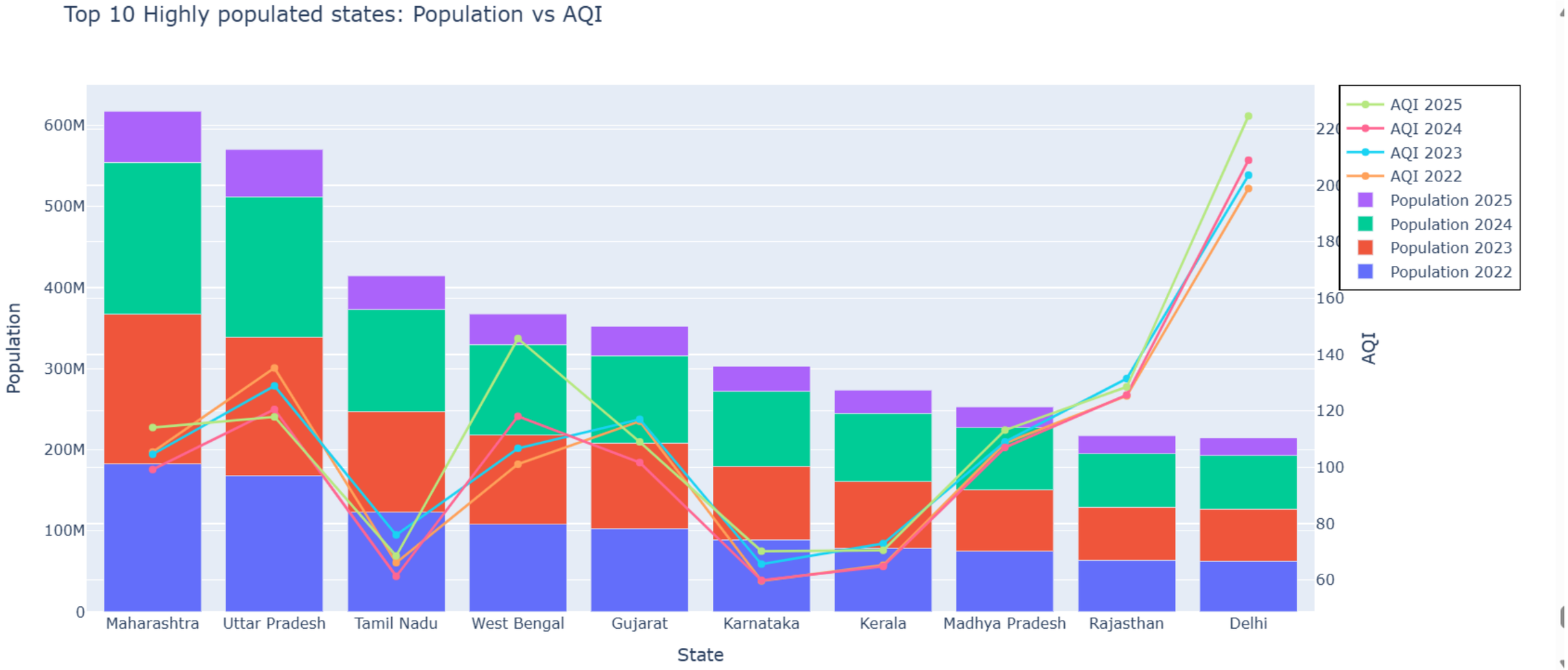
Key Competitors

Brand / Model	Strengths	Common Gaps
Xiaomi Mi Series	Affordable, Smart UI, Fast air cleaning	Weak pre-filter, odor performance, lock-out UX
Coway Airmega 150/Aim	Excellent filtration, long uptime, low noise	Expensive, unstable auto mode, auto-resume issue
Philips / Sharp	Precision sensors, allergen removal	High filter cost, limited coverage/feature set
Dyson	Premium design and features	Very expensive, costly filters

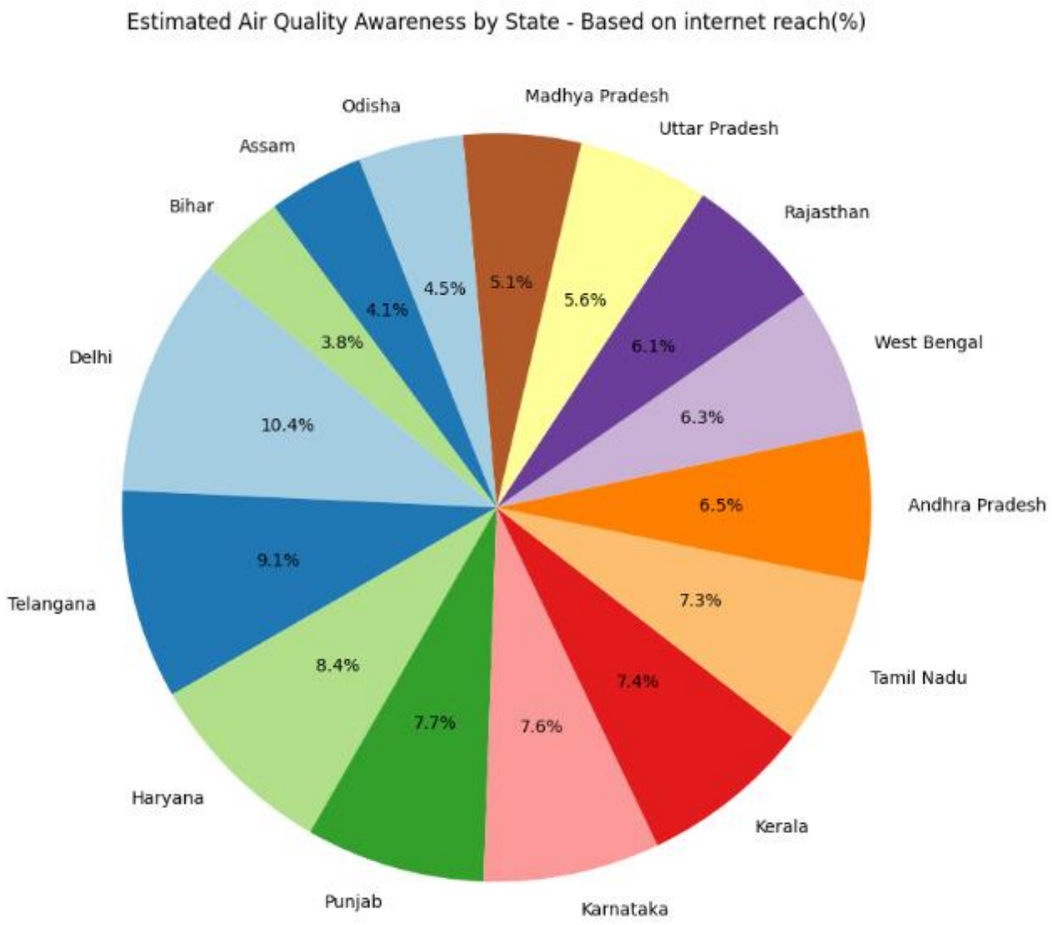
Key differentiators

Feature / Brand	Xiaomi Mi 3/4	Coway AirMega	Philips AC2887	Sharp FP-F40E
CADR (Clean Air Delivery Rate)	350–400 m^3/hr	303 m^3/hr	333 m^3/hr	240 m^3/hr
True HEPA (H13+)	H11-H12	H13	H13	H12
Activated Carbon Filter	Yes	Yes	Yes	Yes
Pre-Filter	Weak/Static	Washable	Washable	Yes
Odor Removal	Average	Moderate	Good	Good
Noise in Sleep Mode	30–35 dB	22–30 dB	33 dB	25–32 dB
Auto Mode Stability	Unstable	Fluctuates	Smooth	OK
Mobile App / Smart	Yes (Mi Home)	No	No	No
PM2.5/PM10 AQI Display	Yes	Yes	Yes	Yes
Filter Life Indicator	Yes (but forced lockout)	Yes	Yes	Yes
Power Resume After Cut	Yes	No	Yes	Yes
Coverage Area	~450 sq.ft.	~350 sq.ft.	~410 sq.ft.	~320 sq.ft.
Filter Replacement Cost	₹2,000–₹3,000	₹4,000+	₹3,000–₹4,000	₹3,000
MRP Range	₹8k–₹13k	₹15k–₹22k	₹13k–₹20k	₹12k–₹17k

Population vs AQI



AQI awareness based on internet reach – Google Trends



Government Initiatives to reduce AQI

Policy	Year	Description	Target Area
National Clean Air Programme (NCAP)	2019	Aim to reduce PM2.5/PM10 by 20-30% by 2024 in 131 cities	Pan-India (non-attainment cities)
BS-VI Emission Norms	2020	Replaced BS-IV with stricter vehicle emissions norms	Nationwide
GRAP (Graded Response Action Plan)	Ongoing (revised 2022)	Emergency measures for Delhi NCR	Delhi NCR
SAFAR Initiative	Expanded post-2020	AQI monitoring & forecasting system	Metro cities
State-specific bans on firecrackers/open burning	Various	Seasonal curbs	Delhi, West Bengal, etc.
Switch to cleaner fuels (e.g., PNG)	Expanded post-2019	Industrial clusters moved to gas	Surat, Ahmedabad, Delhi, etc.

PM2.5/PM10 Pollutants over the years

	Area	PM10_2022	PM10_2023	PM10_2024	PM10_2025	PM2.5_2022	PM2.5_2023	PM2.5_2024	PM2.5_2025	PM25_%_Reduction	PM10_%_Reduction
0	Ahmedabad	95.000000	93.660000	67.960000	93.000000	178.880000	144.840000	142.480000	145.610000	18.600000	2.110000
1	Begusarai	94.520000	206.090000	166.750000	126.300000	283.540000	317.090000	165.600000	181.690000	35.920000	-33.630000
2	Bengaluru	62.520000	66.310000	72.210000	79.580000	128.440000	116.000000	99.500000	104.120000	18.930000	-27.280000
3	Bhopal	117.130000	90.230000	91.150000	115.760000	216.550000	202.920000	207.690000	165.830000	23.420000	1.170000
4	Chennai	70.400000	70.000000	67.860000	71.280000	159.670000	112.630000	111.250000	127.640000	20.060000	-1.250000
5	Delhi	252.670000	156.160000	166.980000	236.610000	313.410000	288.140000	324.550000	282.860000	9.750000	6.350000
6	Ghaziabad	204.400000	154.230000	127.190000	93.250000	287.590000	271.900000	266.920000	215.050000	25.220000	54.380000
7	Gurugram	136.680000	97.400000	104.670000	134.670000	267.480000	215.560000	216.450000	209.740000	21.590000	1.470000
8	Hyderabad	85.430000	80.800000	64.790000	85.340000	116.910000	108.750000	109.360000	103.600000	11.380000	0.100000
9	Indore	108.940000	97.970000	84.960000	85.230000	159.000000	181.750000	127.950000	116.500000	26.730000	21.770000
10	Jaipur	116.130000	103.760000	118.590000	129.160000	189.750000	211.370000	219.660000	184.000000	3.030000	-11.220000
11	Kolkata	70.250000	63.910000	80.390000	79.410000	235.690000	184.280000	191.510000	160.190000	32.040000	-13.040000
12	Mumbai	107.500000	88.020000	60.460000	87.310000	271.000000	154.670000	150.650000	134.380000	50.420000	18.780000
13	Nagpur	71.240000	75.820000	73.780000	100.600000	180.960000	166.460000	171.770000	170.650000	5.690000	-41.210000
14	Pune	221.670000	116.500000	55.330000	105.000000	133.650000	166.220000	155.230000	137.180000	-2.650000	52.630000

Month	Theme / Mission	Key Activities & Description
May	Plastic-Free Public Places	Clean-up drives, promoting eco-friendly alternatives to single-use plastics (EduDEL, NCR Guide)
June	Plant Trees, Breathe Free	"Ek Ped Maa Ke Naam 2.0" plantation drives and World Environment Day events (June 5) (The Times of India, NCR Guide)
July	Reduce Industrial Emissions for Blue Skies	Visits to clean-tech industries, advocacy for LPG/PNG in commercial establishments (The Times of India, EduDEL)
August	Clean Air Heroes – Be One	Celebrate individuals/groups working for clean air via street plays, storytelling sessions (The Times of India, NCR Guide)
Sept	Every Breath Matters	Week-long awareness leading up to Swachh Vayu Diwas; include Zero Emission Day pledges, art installations in public spaces (The Times of India, EduDEL)
October	Burn Calories, Not Fuel	Encourage non-motorized travel (walking/cycling), engine-off campaigns at red lights (The Times of India, NCR Guide)
November	Green Living, Better Tomorrow	Poster/essay contests, short film screenings, "Power of Change" activities in communities (The Times of India, NCR Guide)
December	Sustainable Cities, Healthier Air	Promote green urban planning, sustainable construction and open-space design (The Times of India, EduDEL)
January	Say No to Waste Burning	Anti-waste-burning campaigns—public pledges, neighborhood clean-ups, advocacy to avoid open burning (The Times of India, NCR Guide)
February	Reduce, Reuse, Repurpose, Recycle	Visits to recycling units, zero-waste fairs, reuse challenges, circular economy awareness (The Times of India, NCR Guide)
March	No More Dust	Street cleaning, dust-control drives, neighborhood campaigns to reduce particulate pollution (The Times of India, NCR Guide)

Cities with irreversible AQI degradation

(High average AQI for past 3 years)

City	category	avg_aqi_2022_2025
Delhi	Tier 1	206.4168889
Ghaziabad	Tier 2	183.0106762
Patna	Tier 2	179.9271111
Meerut	Tier 2	158.3505798
Chandigarh	Tier 2	141.5617778

Pediatric Asthma vs AQI

Location / Study	Pollutant & AQI Spike	Health Outcome
Mysore (time-series) (PubMed)	Daily increases in **NO_2** & **$PM_{2.5}$**	Hospital admissions for asthma & wheezing
Delhi ER mapping (Mar 2018–Feb 2019) (NCBI)	Seasonal high **$PM_{2.5}$** in winter ("very poor" AQI days)	ER visits for acute pediatric respiratory symptoms
Delhi traffic-exposure cohort (Lippincott Journals)	Residential proximity to traffic (**NO_2**, PM_{10})	Asthma exacerbations in children with existing asthma
Delhi vs Kottayam/Mysore survey (PMC)	Chronic ambient PM_{10}/$PM_{2.5}$ (Delhi extremely polluted)	Asthma prevalence and symptoms
National burden (vehicle NO_2) (researchmatters.in, Hindustan Times)	Traffic-related NO_2 exposure	New pediatric asthma cases

Pollution emergencies vs purifier purchases

Sales Spike Triggers	- Seasonal surges during **late autumn & winter** (esp. Oct–Nov) - Spikes linked to **Diwali, stubble burning, AQI > 300**
Spike Examples	- **Q3 2023**: 64% YoY national surge - Diwali weeks: up to **10x** sales in metros - 4–5 day windows saw **15–20%** demand spikes
Brand Insights	- Kent, Xiaomi, LG, Philips report up to **50% increase** when AQI worsens - 200% QoQ rise for specific models (Philips, Dyson)
Delhi NCR Trends	- 80% of commercial buildings now use purifiers - Sales rise up to 50% in peak pollution season
Consumer Behavior	- Highly **event/news-driven** buying behavior - **Media coverage** and **AQI headlines** directly influence purchases
Correlation with AQI	- Strong, direct link between **AQI spikes** and **sales/search interest**

Health burden projections – Delhi 2025 case study

Scenario	AQI Impact	Pediatric Pop. (6–14 yrs)	Asthma Prevalence (%)	# Asthmatic Children	Hospitalization Rate (%)	Est. Hospitalizations	Monthly Cost/Child (₹)	Total Direct Monthly Cost (₹)
Baseline (Clean Air)	AQI < 100 (WHO safe)	2,500,000	8%	200,000	2%	4,000	₹410	₹8.2 Cr
Current (Avg AQI ~250)	Moderate–high AQI days common	2,500,000	30%	750,000	6%	45,000	₹410	₹30.75 Cr
Severe Spike (AQI > 400)	Post-Diwali, winter AQI crisis	2,500,000	35%	875,000	8%	70,000	₹500 (incl. ER visits)	₹43.75 Cr

Competitor Feature Gap Matrix

Feature	Xiaomi Mi 3/4	Coway AirMega	Philips AC2887	Sharp FP-F40E
CADR ≥ 400 m^3/hr	❌	❌	❌	❌
True HEPA (H13 or better)	❌ (H11–H12)	✅ (H13)	✅ (H13)	❌ (H12)
Activated Carbon Filter	✅	✅	✅	✅
VOC Removal	❌	❌	❌	❌
Pre-Filter (Washable & Robust)	❌ (Static)	✅ (Washable)	✅ (Washable)	❌ (Basic)
Odor Removal (Excellent)	❌	❌	✅	✅
Sleep Mode Noise <25 dB	❌ (30–35 dB)	✅ (~22–30 dB)	❌ (33 dB)	❌ (25–32 dB)
Stable Auto Mode	❌	❌	✅	❌
Mobile App / Smart Connectivity	✅	❌	❌	❌
AQI Display (PM2.5/PM10)	✅	✅	✅	✅
Filter Life Indicator	✅	✅	✅	✅
No Forced Lockout on Filters	❌	✅	✅	✅
Auto Power Resume	✅	❌	✅	✅
Coverage ≥ 500 sq.ft.	❌ (~450)	❌ (~350)	❌ (~410)	❌ (~320)
Filter Replacement Cost < ₹2,500	❌ (₹2k–₹3k)	❌ (₹4k+)	❌ (₹3k–₹4k)	❌ (~₹3k)
Price within ₹10k–₹14k	✅	❌	❌	✅

Feature Comparison Matrix (with Our Product)

Feature	Xiaomi Mi 3/4	Coway AirMega	Philips AC2887	Sharp FP-F40E	⭐ Our Product
CADR ≥ 400 m^3/hr	❌	❌	❌	❌	✅
True HEPA (H13 or better)	❌ (H11–H12)	✅ (H13)	✅ (H13)	❌ (H12)	✅ (H13–H14)
Activated Carbon Filter	✅	✅	✅	✅	✅ + VOC
VOC Removal	❌	❌	❌	❌	✅
Pre-Filter (Washable & Robust)	❌ (Static)	✅ (Washable)	✅ (Washable)	❌ (Basic)	✅
Odor Removal (Excellent)	❌	❌	✅	✅	✅
Sleep Mode Noise <25 dB	❌ (30–35 dB)	✅ (~22–30 dB)	❌ (33 dB)	❌ (25–32 dB)	✅ (<25 dB)
Stable Auto Mode	❌	❌	✅	❌	✅ (AI-based)
Mobile App / Smart Connectivity	✅	❌	❌	❌	✅ (AQI + Remote)
AQI Display (PM2.5/PM10)	✅	✅	✅	✅	✅ + Outdoor Sync
Filter Life Indicator	✅	✅	✅	✅	✅ (No Lockout)
No Forced Lockout on Filters	❌	✅	✅	✅	✅
Auto Power Resume	✅	❌	✅	✅	✅
Coverage ≥ 500 sq.ft.	❌ (~450)	❌ (~350)	❌ (~410)	❌ (~320)	✅ (>500)
Filter Replacement Cost < ₹2,500	❌ (₹2k–₹3k)	❌ (₹4k+)	❌ (₹3k–₹4k)	❌ (~₹3k)	✅
Price within ₹10k–₹14k	✅	❌	❌	✅	✅

What Sets “Our Product” Apart

Feature / Brand	Xiaomi Mi 3/4	Coway AirMega	Philips AC2887	Sharp FP-F40E	Our Product
CADR (Clean Air Delivery Rate)	350–400 m^3/hr	303 m^3/hr	333 m^3/hr	240 m^3/hr	400+ m^3/hr
True HEPA (H13+)	H11-H12	H13	H13	H12	H13–H14
Activated Carbon Filter	Yes	Yes	Yes	Yes	Yes + VOC
Pre-Filter	Weak/Static	Washable	Washable	Yes	Robust + Washable
Odor Removal	Average	Moderate	Good	Good	Excellent (VOC + Ozone Free)
Noise in Sleep Mode	30–35 dB	22–30 dB	33 dB	25–32 dB	<25 dB
Auto Mode Stability	Unstable	Fluctuates	Smooth	OK	Stable AI-based
Mobile App / Smart	Yes (Mi Home)	No	No	No	Yes + Real-time AQI + Remote
PM2.5/PM10 AQI Display	Yes	Yes	Yes	Yes	Yes + Outdoor Sync
Filter Life Indicator	Yes (but forced lockout)	Yes	Yes	Yes	Yes – No Lockout
Power Resume After Cut	Yes	No	Yes	Yes	Yes
Coverage Area	~450 sq.ft.	~350 sq.ft.	~410 sq.ft.	~320 sq.ft.	500+ sq.ft.
Filter Replacement Cost	₹2,000–₹3,000	₹4,000+	₹3,000–₹4,000	₹3,000	< ₹2,500
MRP Range	₹8k–₹13k	₹15k–₹22k	₹13k–₹20k	₹12k–₹17k	₹6k–₹16k

Target Cities for launch based on AQI , Population & Income

City Risk Score = (0.5×AQI + 0.3×Population + 0.2 x Income)

Category	City	AQI_Rank	Population (2025	PCI (in lakhs)	Risk_score	Major issues
Tier 1	Delhi	Very High	34.7M	4.61	0.91	Highest AQI in the world at times
	Mumbai	High	22.1M	4.13	0.45	Coastal, but construction & traffic
	Bengaluru	Moderate	14.4M	7.6	0.39	Tech hub, rising construction dust
	Kolkata	High	15.8M	1.54	0.30	Old industries, coal, traffic
	Hyderabad	Moderate	11.3M	4.54	0.29	Expanding metro with rising AQI
	Chennai	Moderate	12.3M	1.96	0.19	Industrial + vehicular pollution
Tier 2	Ghaziabad	High	1.2M	1.35	0.43	road dust, waste burning, traffic
	Patna	High	2.7M	1.2	0.43	Brick kilns + traffic
	Chandigarh	High	1.3M	4	0.38	Seasonal pollution
	Meerut	High	1.9M	1.3	0.36	waste burning
	Ahmedabad	High	9.1M	3.89	0.35	Textile & chemical zones
	Pune	High	7.5M	2.78	0.30	Tier-1 for consumer tech & health
	Jaipur	High	4.4M	1.5	0.30	AQI fluctuates, especially winter
	Surat	High	8.6M	1.8	0.29	Textile industries
	Lucknow	High	4.1M	1.25	0.29	AQI often >200, seasonal smog
	Nagpur	High	3.2M	2.78	0.26	Central India hub, dry dust
	Bhopal	High	2.7M	1.43	0.24	Expanding rapidly, moderate AQI
	Visakhapatnam	High	2.4M	1.6	0.23	Port city + refineries

Recommendations



Use air purifiers



Avoid outdoor exposure on bad AQI days



Switch to clean fuels



Ban crop and waste burning



Promote EVs and green spaces



Thank you

- Air pollution is a daily health threat.
- Protecting future generations begins with action today.